

Guidelines for Renewable Energy Integrated Power System Operation and Dispatch

This document provides guidelines for the basic operation and dispatch of a power system, outlining a comprehensive process to achieve optimal dispatch schemes for power system operators. The operational workflow mainly includes data acquisition, forecasting of load and renewable energy production, and the optimization of power dispatch strategies. A dynamic alternating current (AC) optimal power flow (OPF) approach, based on 15-minute dispatch intervals, is specifically adopted to ensure real-time responsiveness and efficiency.

Data acquisition is crucial for forecasting load and renewable energy generation. It involves retrieving historical load data and renewable energy generation data from the supervisory control and data acquisition (SCADA) system. Moreover, given the close relationship of these forecasts with weather parameters, relevant future weather data (e.g., wind speed, solar irradiation, temperature, humidity) are also obtained from external weather forecasting systems via established interfaces and protocols. The forecasting process involves predicting the load and renewable energy generation at different buses of the power system 15 minutes ahead. These forecast models, based on cutting-edge deep learning techniques, have been well-trained for practical application. Operators are required to input the acquired data into these forecast models, thus obtaining prediction results for load and renewable energy generation at each bus of the power system 15 minutes into the future.

The dispatch process utilizes the AC OPF optimization model to produce a comprehensive dispatch plan that specifies the required output from each unit 15 minutes later. This AC OPF model incorporates various factors (e.g., unit costs, renewable energy curtailment costs, forecasted load and renewable energy generation, power balance, transmission line constraints, unit ramping constraints, and so on), enabling the delivery of an optimal dispatch plan. Operators need to input these data into the AC OPF model to derive the detailed dispatch scheme for each unit. The overall operation and dispatch process is a routine task that is conducted and repeated every 15 minutes, as outlined by the following steps.

1. Obtain relevant forecast data from CAISO, which can be done by running “data_loader.py”, such that data required for the next 15-minute interval prediction is prepared.
2. Forecast load and renewable energy generation at each bus, which can be done by transmitting relevant data to these forecast models and running “predict.py”. The forecasted results are stored in CSV format.
3. Perform AC OPF to yield an optimal dispatch plan for each unit, which can be done by loading the forecasted results and running the Matlab script “dispatch.m” by the command line “matlab -batch ‘dispatch’”. At last, the detailed dispatch plan for each unit will be stored in a folder named after the target time point.